

## Chapter 2

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## Demand functions:

The consumer's choice depends on prices  $P$  and wealth or income, we let  $x(P, w)$  denote the demand correspondence. The set of bundles that are acceptable to consumer. If correspondence is single-valued

- The set consists of a single element then we have

A standard demand function. In the latter case

$$x(P, w) = (x_1(P, w) \dots x_L(P, w))$$

$x_i(P, w)$  is the individual demand for good  $i = 1, 2, \dots, L$

We start with some basic assumptions to the effect that consumer is rational

$x(P, w)$  is Homogeneous of degree 0.

$x(\alpha P, \alpha w) = x(P, w)$  For all  $\alpha > 0$ . This is saying that only real price and real income matter. It does not matter if we convert to different currency or from dollar to cents

Walra's Law For any  $P \gg 0$  and  $w > 0$ , the

$$P \cdot x(P, w) = w$$

agent uses all his wealth on income.

We now assume that the demand correspondence is a demand function. We often assume it's differentiable.

Comparative statics

The parameters are  $(P, w)$ . What happens to endogenous choices (i.e. consumption) when exogenous parameters change.

With differentiable demand functions We want to examine

$$\frac{\partial x_i(P, w)}{\partial P_i} \text{ and } \frac{\partial x_i(P, w)}{\partial w}$$

We now use assumptions for comparative statics using homogeneity degree zero

$$x(\alpha P, \alpha w) = x(P, w) \text{ for } \alpha > 0$$

$$\frac{\partial x_i(\alpha P, \alpha w)}{\partial \alpha} = 0$$

$$= \frac{\partial x_i(\alpha P_1, \dots, \alpha P_L)}{\partial \alpha}$$

$$0 = \sum_{i=1}^I \frac{\partial x_i(p, w)}{\partial p_i} p_i + \frac{\partial x_i(p, w)}{\partial w} w$$

evaluate at ~~alpha~~  $\alpha = 1$

$$0 = \sum_{j=1}^I \frac{\partial x_i(p, w)}{\partial p_j} p_j + \frac{\partial x_i(p, w)}{\partial w} w$$

When  $x_i(p, w) > 0$

$$x_i(p, w) \left( \sum_{i=1}^I \frac{\partial x_i(p, w)}{\partial p_i} \frac{p_i}{x_i(p, w)} + \frac{\partial x_i(p, w)}{\partial w} \frac{w}{x_i(p, w)} \right)$$

All income elasticities sum to zero.

This is intuitive given the starting point is homogeneity of 0 degree. Nothing changes if all prices and wealth change by the same percent. We learn that some elasticities are + and some negative.

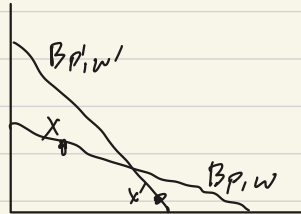
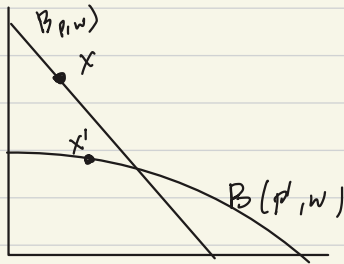
The weak axioms of revealed preference, (WARP) ~~minimize~~ minimal rationality assumptions.

The agent doesn't make self-contradictory choices.

The WARP is satisfied if for any two  
 $(p, w)$  and  $(p', w')$ . It holds that

if  $p \cdot x(p', w') \leq w$  Tom. bundle is affordable today  
 $x(p', w') \neq x(p, w)$  ... but I chose not to buy it.  $x(p, w)$  is revealed  
 $p' \cdot x(p, w) > w'$  pref to  $x(p', w')$

The reason I don't buy  $x(p, w)$  tom. is simply I cannot afford it.



WARP fails.

WARP implies homogeneity of degree zero.

If  $(p', w') \sim (p, w)$  Then  $x'$  is affordable when  $x$  was chosen as vice versa.

If  $x' \neq x$ . Then  $x$  is r.p. to  $x'$  and  $x'$  is R.P. to  $x$   
- A self-contradiction choice that violate WARP.  $\Rightarrow x' = x$

WARP has strong implications when combined with Walra's Law

A thought experiment: price change from  $p$  to  $p'$ . At the same time, adjust income so

$w' = p' \cdot x(p, w) \rightarrow$  Slutsky wealth compensation.

Then compensation of demand assume  $x(p, w)$   
satisfies Walra's Law

Then  $x(p, w)$  satisfies WARP if and only if

for any  $(p', w')$  with  $w' = p' \cdot x(p, w)$ . It holds

that  $(p' - p)(x(p', w)' - x(p, w)) = 0$

strict if  $x(p, w) \neq x(p', w')$

In aggregate case, demand move in opposite direction  
to prices

$$(p_i' - p_i)(x_i(p_i', w') - x_i(p_i, w)) \leq 0$$

→ demand for good moves in opposite dire.  
at its price.

